F78AB – Actuarial and Financial Mathematics
Computer Lab 2

Question 1:

At the end of every year, for 10 years, a payment of \$2,000 is paid into an account, Interest is paid following a constant force of interest of 0.04 per annum. Using Excel, show that the total present value of the payments is \$16,156.52.

Good practice would mean you should do the following.

i) Show the payment made in each year,
ii) Calculate the present value, at time 0, of each payment.
iii) Sum these individual present values to give the total present value of this stream of payments.

Question 2:

Payments to contractors on a housing project are made monthly in advance for 15 months and increase by 3% per month effective. The initial monthly payment is \$4 million.

The constant force of interest is 1% per month.

Calculate the total present value of the cashflows.

Answer: \$69.050 million.

Question 3:

For this question, time is measured in years.

At time $t = 0$, an investor believes that the annual force of interest $\delta(t)$ on a savings account will be

$$\delta(t) = \begin{cases} 0.03 \text{ p.a.} & \text{for } 0 \leq t \leq 4 \\ 0.05 \text{ p.a.} & \text{for } 4 < t. \end{cases}$$

The investor plans to deposit £10 000 in the savings account at the end of every year for the next 20 years.

Calculate the total present value of the cashflows.

Answer: \$132,389.24.

Question 4:

Use Excel to do the calculations for **example 5.1.1** from the notes.

Remember to use as little hardcoding as possible.

The Excel file (RPI Data 1987-2014.xlsx) has data on the RPI for January 1987 to December 2014.

Question 5:

Do the calculations for **Tutorial 3 Question 4.1** in Excel.

Question 6:

Do the calculations for **Tutorial 3 Question 5.1 and 5.2** in Excel.